MECH5200 Numerical Methods for PDEs : *Video 9: 2D Finite Difference Equations*

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(Please also see MIT's OCW 16.920 Notes)

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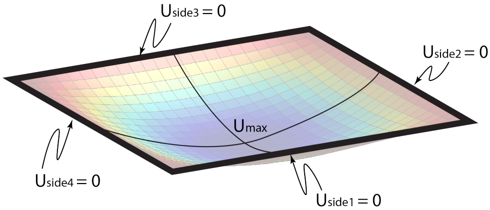
Thought Experiment

- Let's extend the string equation to a membrane:

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y) \quad (1)$$



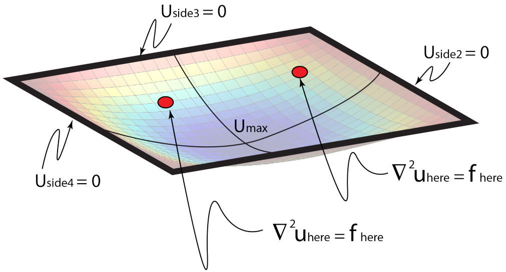
- With boundary conditions on all sides:

$$u_{side_1} = u_{side_2} = u_{side_3} = u_{side_4} = 0 \quad (2)$$

Figure: The membrane problem domain.

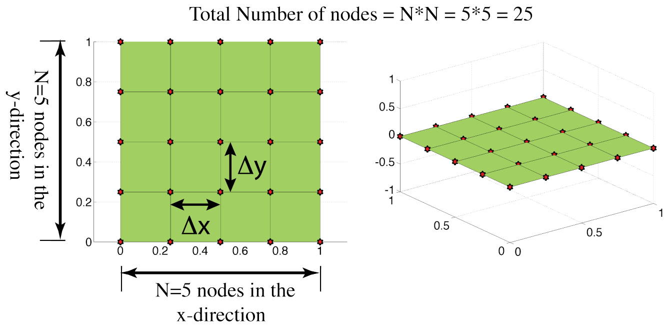
Thought Experiment

- **Goal:** Satisfy the governing equation at each location on the membrane



Step 1: Discretization

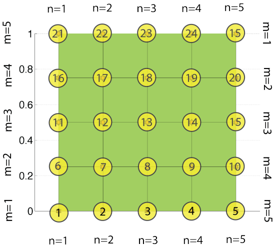
- Break membrane into a series of equally sized "chunks".
- Let's break the membrane into N nodes in the $x-$direction and M nodes in the $y-$direction:

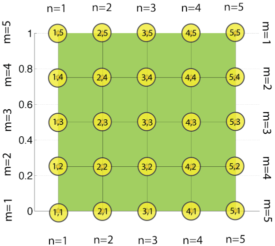


- Total number of nodes $= N * M$.
- Note, when: $\Delta x = \Delta y \rightarrow N = M$ for a square domain (simplifying things).

Step 1: Discretization

- Recall, we would like to satisfy the PDE at each node
  - 1 x equation per node
- To simplify life, let's number the nodes as:





- Relating the node number (NN) to the n and m index (n, m):

$$NN = N * (m - 1) + n$$

- This node numbering scheme will become quite useful when we build the matrix (later).

Step 2: Discretize the Governing Equation using FD Approximations of Derivatives

- **Governing equation**: At each node, i, of the domain, we want to specify the equation:

$$\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)_i = f_i(x, y)$$

- We can write the discrete finite difference equation for node (n, m):

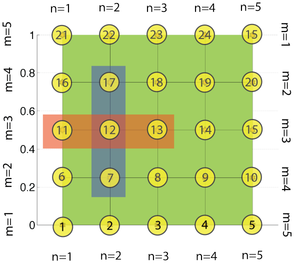
$$\frac{u_{n-1} - 2u_n + u_{n+1}}{(\Delta x)^2} + \frac{u_{m-1} - 2u_m + u_{m+1}}{(\Delta y)^2} \simeq f_{n,m}(x, y)$$

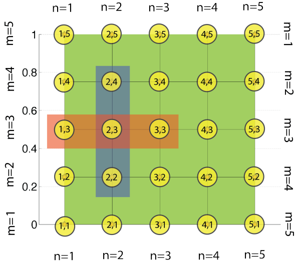
- **Finite Differences:** Determine the solution u that satisfies the (approximate) finite difference equations at each of the nodes of the discrete domain.

Step 2: Discretize the Governing Equation

- Let's look at **i = node 12** to begin (n = 2, m = 3):

$$\left(\frac{u_{n-1,m} - 2u_{n,m} + u_{n+1,m}}{(\Delta x)^2}\right) + \left(\frac{u_{n,m-1} - 2u_{n,m} + u_{n,m+1}}{(\Delta y)^2}\right) \simeq f_{n,m}$$





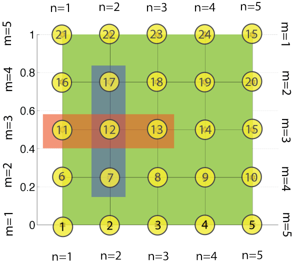
$$\frac{u_{1,3} - 2u_{2,3} + u_{3,3}}{(\Delta x)^2} + \frac{u_{2,2} - 2u_{2,3} + u_{2,4}}{(\Delta y)^2} \simeq f_{2,3}$$

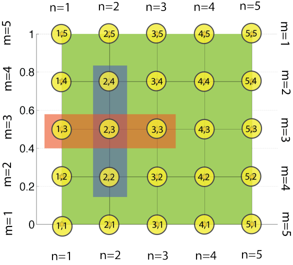
Step 2: Discretize the Governing Equation

- So, we have this:

$$\frac{u_{1,3} - 2u_{2,3} + u_{3,3}}{(\Delta x)^2} + \frac{u_{2,2} - 2u_{2,3} + u_{2,4}}{(\Delta y)^2} \simeq f_{2,3}$$

- But, now consider how the equation at **node 12** can be written using the i-node numbers (not m, n locations) :

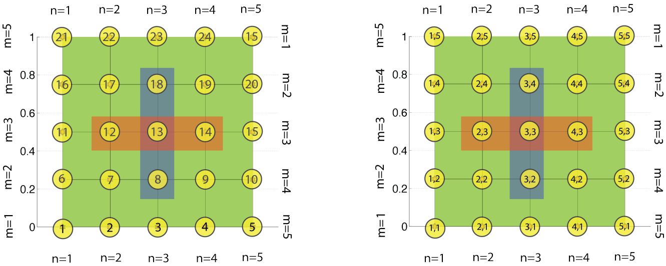




$$\frac{u_{11} - 2u_{12} + u_{13}}{(\Delta x)^2} + \frac{u_7 - 2u_{12} + u_{17}}{(\Delta y)^2} \simeq f_{12}$$

Step 2: Discretize the Governing Equation

- **Let's repeat the same analysis for node 13:**

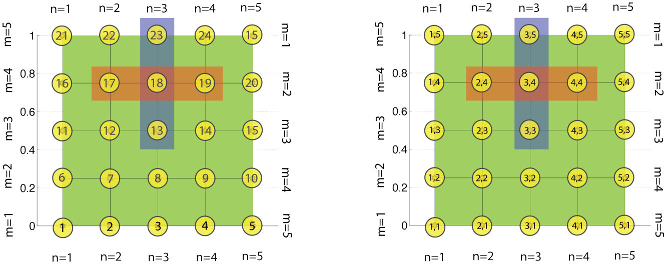


- So, for node 13, the equation *approximating the PDE* is:

$$\textit{Node}\ \ 13: \qquad \frac{u_{12} + u_{14} + u_{8} + u_{18} - 4u_{13}}{(\Delta x)^2} \simeq f_{13}$$

Step 2: Discretize the Governing Equation

- **Let's repeat the same for node 18:**



- So, for node 18, the equation *approximating the PDE* is:

$$\textit{Node}\ \ 18: \qquad \frac{u_{17} + u_{19} + u_{13} + u_{23} - 4u_{18}}{(\Delta x)^2} \simeq f_{18}$$

ASIDE: Using the Finite Differences in 2D to Understand the PDE Operator

- So, according to the node numbering scheme used here, the equation **approximating the PDE** at **node 12** is:

$$\frac{u_{11} + u_{13} + u_7 + u_{17} - 4u_{12}}{(\Delta x)^2} \simeq f_{12}$$

ASIDE: A moment to understand the meaning of $\nabla^2 u = 0$

- Consider when f_{12} is zero:

$$\frac{u_{11} + u_{13} + u_7 + u_{17} - 4u_{12}}{(\Delta x)^2} \simeq 0$$

- Rearrange to describe u_{12}:

$$\frac{u_{11} + u_{13} + u_7 + u_{17}}{4} \simeq u_{12}$$

- The value of u_{12} looks a lot like it is taking the "average" value of the 4 x surrounding u-values.

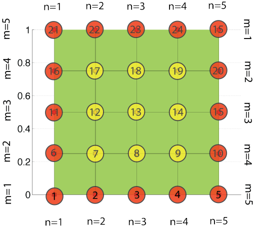
Step 2: Discretize the Governing Equation

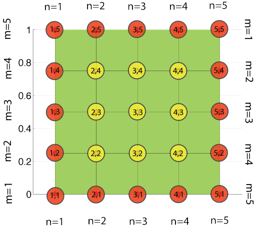
- **Summary**: for the 3 x internal nodes we looked at
- Note: the same approach generalizes for other internal nodes.

$$\textit{Node}\ \ 12: \qquad \frac{u_{11} + u_{13} + u_{7} + u_{17} - 4u_{12}}{(\Delta x)^2} \simeq f_{12}$$

$$\textit{Node}\ \ 13: \qquad \frac{u_{12} + u_{14} + u_{8} + u_{18} - 4u_{13}}{(\Delta x)^2} \simeq f_{13}$$

$$\textit{Node}\ \ 18: \qquad \frac{u_{17} + u_{19} + u_{13} + u_{23} - 4u_{18}}{(\Delta x)^2} \simeq f_{18}$$





Step 3: Form a Linear System of Equations

- Each node's finite difference equation has 5-unknowns represented.
- For node 12 the equation we found is:

$$Node\ \ 12: \qquad \frac{1}{(\Delta x)^2}\left(u_{11} + u_{13} + u_7 + u_{17} - 4u_{12}\right) \simeq f_{12}$$

- We can re-write this equation to show that most of the unknowns have coefficients of zero in this equation:

$$\frac{1}{(\Delta x)^2} * [0u_1 + ... + 1 * u_7 + ... + 1 * u_{11} - 4 * u_{12} + 1 * u_{13} + ... + 1 * u_{17} + ...] = f_{12}$$

- Once we start to consider the equation from each node, we end up with a system of equations!

Step 3: Form a Linear System of Equations

- For node 12, we can write the equation as a single row of a Matrix system:

$$\textit{Node}\ \ 12: \qquad \frac{1}{(\Delta x)^2}\left(u_{11} + u_{13} + u_7 + u_{17} - 4u_{12}\right) \simeq f_{12}$$

$$\frac{1}{(\Delta x)^2}\begin{bmatrix} \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \vdots & 1 & \ldots 0 \ldots & 1 & -4 & 1 & \ldots 0 \ldots & 1 & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix} \begin{pmatrix} u_1 \\ \vdots \\ u_7 \\ \vdots \\ u_{11} \\ u_{12} \\ u_{13} \\ \vdots \\ u_{17} \\ \vdots \\ u_N \end{pmatrix} = \begin{pmatrix} \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ f_{12} \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ u_N \end{pmatrix}$$

Step 3: Form a Linear System of Equations

$$Node\ \ 13: \qquad \frac{1}{(\Delta x)^2}\left(u_{12} + u_{15} + u_8 + u_{18} - 4u_{13}\right) \simeq f_{13}$$

$$\frac{1}{(\Delta x)^2}\begin{bmatrix}
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
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\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
0 & \vdots & 1 & \ldots 0 \ldots & 1 & -4 & 1 & \ldots 0 \ldots & 1 & \vdots & \vdots \\
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\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots
\end{bmatrix}
\begin{pmatrix} u_1 \\ \vdots \\ u_8 \\ \vdots \\ u_{12} \\ u_{13} \\ u_{14} \\ \vdots \\ u_{18} \\ \vdots \\ u_N \end{pmatrix}
=
\begin{pmatrix} \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ f_{13} \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ u_N \end{pmatrix}$$

Step 3: Form a Linear System of Equations

$$\textit{Node}\ \ 18: \qquad \frac{1}{(\Delta x)^2}\left(u_{17} + u_{19} + u_{13} + u_{23} - 4u_{18}\right) \simeq f_{18}$$

$$\frac{1}{(\Delta x)^2}\begin{bmatrix}
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
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0 & \vdots & 1 & \ldots 0 \ldots & 1 & -4 & 1 & \ldots 0 \ldots & 1 & \vdots & \vdots \\
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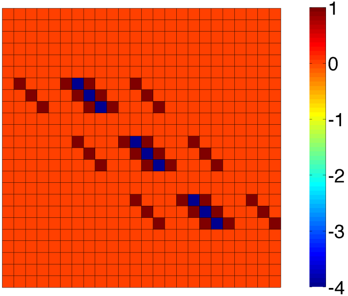
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots
\end{bmatrix}
\begin{pmatrix} u_1 \\ \vdots \\ u_{13} \\ \vdots \\ u_{17} \\ u_{18} \\ u_{19} \\ \vdots \\ u_{23} \\ \vdots \\ u_N \end{pmatrix}
=
\begin{pmatrix} f_1 \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ f_{18} \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ u_N \end{pmatrix}$$

Step 3: Form a Linear System of Equations

- This results in a **linear system of equations**, $Au = f$
- Looking only at the A-matrix for right now:

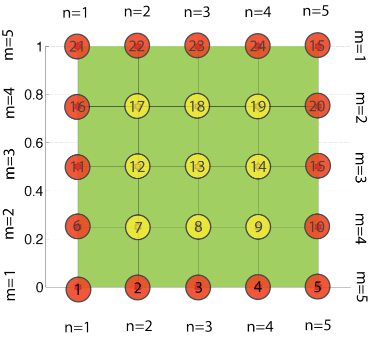


- Examining the Structure of A (for internal nodes):
  - The value -4 is on the diagonal entry for equations with internal nodes
  - The value 1 is in the off diagonal entries

Step 3: Form a Linear System of Equations

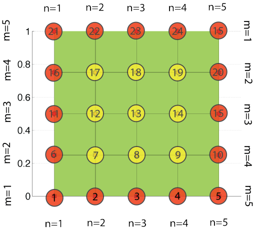
- Don't forget to setup the RHS vector
- In this example $f = 1$ at each of the internal nodes

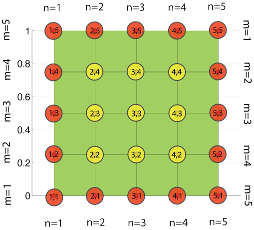
$$f = \begin{pmatrix} f_1 \\ \vdots \\ f_6 \\ f_7 \\ f_8 \\ f_9 \\ f_{10} \\ f_{11} \\ f_{12} \\ f_{13} \\ f_{14} \\ f_{15} \\ \vdots \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 1 \\ 1 \\ 1 \\ 0 \\ \vdots \end{pmatrix}$$



Step 4: Incorporate Boundary Conditions

- Introduce boundary conditions.
- We know that $u = 0$ at all boundary nodes

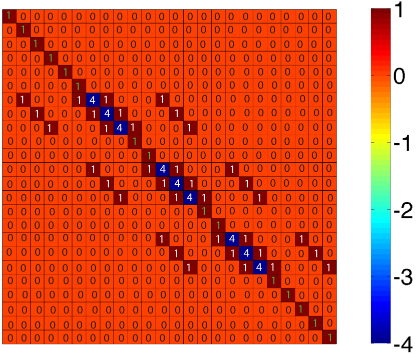




$$
\begin{aligned}
\textit{bottom}: \quad u_1 = u_2 = u_3 = u_4 = u_5 &= 0 \\
\textit{top}: \quad u_{21} = u_{22} = u_{23} = u_{24} = u_{25} &= 0 \\
\textit{left}: \quad u_6 = u_{11} = u_{16} &= 0 \\
\textit{right}: \quad u_{10} = u_{15} = u_{20} &= 0
\end{aligned}
$$

Step 4: Incorporate Boundary Conditions

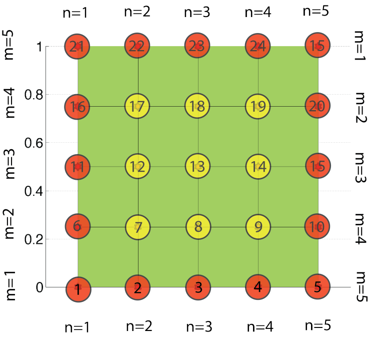
- Need to incorporate these boundary conditions into A:
  - **Example:** $u_{22} = 0 = f_{22}$:
    - Make the 22nd-row of the A-matrix $= 0$
    - Insert a value of Δx^2 on the diagonal ($A_{22,22} = dx^2$)
    - Set the 22nd-entry of the f-vector $= 0$ ($f_{22} = 0$)
  - When applied to all boundary conditions, the A-Matrix takes the form:



Step 4: RHS Vector Boundary Nodes

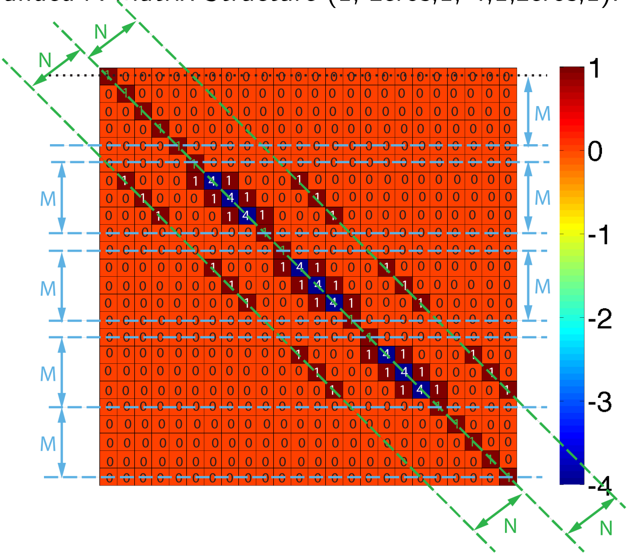
- Don't forget to setup the RHS vector
- In this example $RHS = 0$ at each of the **boundary nodes**

$$f = \begin{pmatrix} f_1 \\ \vdots \\ f_6 \\ f_7 \\ f_8 \\ f_9 \\ f_{10} \\ f_{11} \\ f_{12} \\ f_{13} \\ f_{14} \\ f_{15} \\ \vdots \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 1 \\ 1 \\ 1 \\ 0 \\ \vdots \end{pmatrix}$$



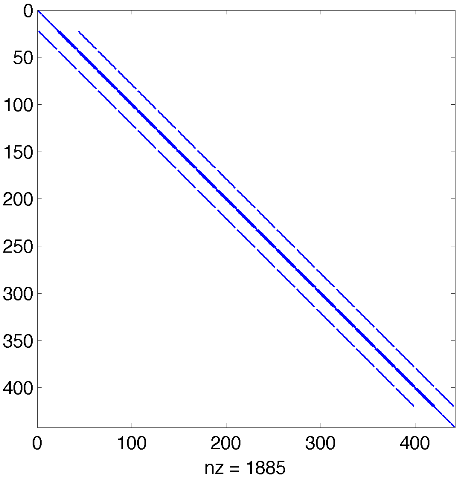
Examine the A-Matrix Structure

- The banded A-Matrix Structure (1, zeros,1,-4,1,zeros,1):



Examine the A-Matrix Structure

- The A-Matrix Structure (21 x 21 nodes: Use, **spy(A)**):



Solve using MATLAB

- Solve the linear system of equations using MATLAB.
- We will use the 'backslash' operator in MATLAB:

$$Solution = A \backslash f \tag{3}$$

Solution

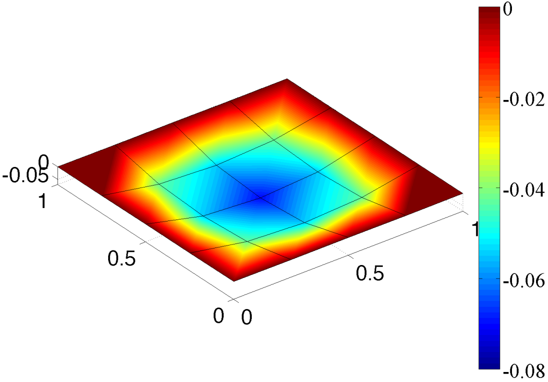


Figure: Solution for $N = M = 5, NumNodes = 25$, Max Deflection = -0.06667

Solution

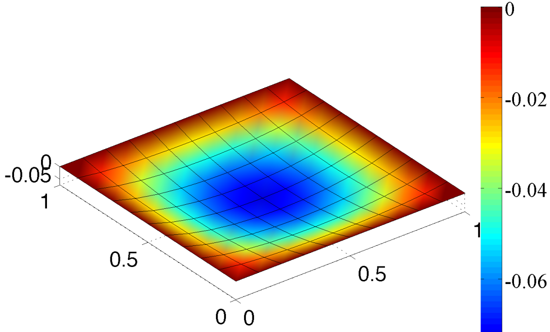


Figure: Solution for $N = M = 11, NumNodes = 121$, Max Deflection = -0.073010

Solution

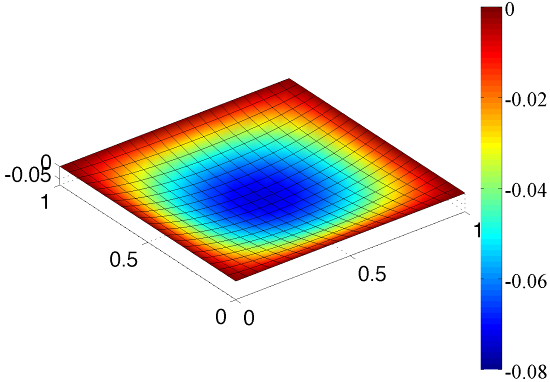


Figure: Solution for $N = M = 21, NumNodes = 441$, Max Deflection = -0.07353

Solution

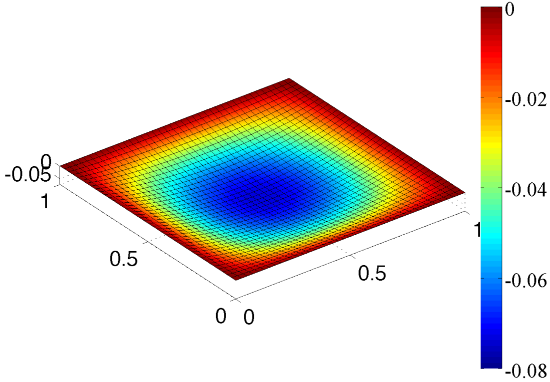


Figure: Solution for $N = M = 41$, $NumNodes = 1681$, Max Deflection = -0.07364

Starting to think about the Error

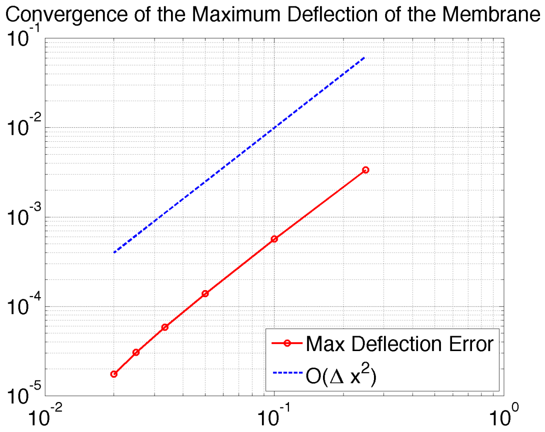


Figure: Convergence of the Max Deflection Value (compared with deflection for N = M = 101)

What have we learned?

- Finite difference expressions in 2D follow the same structure as 1D:
  - Approximate the PDE expression at nodes in the domain
  - Determine the solution at those nodes
- The general steps for a finite difference problem are:
  - **Step 1:** Discretize the domain (in 1D: $N-1$ intervals, N points).
  - **Step 2:** Write finite difference equations that approximate governing equation at each internal node of the problem
  - **Step 3:** Form a linear system ($Ax = b$)
  - **Step 4:** Enforce the boundary conditions where relevant
  - **Step 5:** Solve the system of equations